



Math 4641: Numerical Methods

Lecture 18 — LU Decomposition, Eigenvalues, and Eigenvectors

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Lecture Plan

The agenda for today

- Recap the concept of LU Decomposition from Linear Algebra
- Find the LU Decomposition of an example matrix
- Recap the concept of Eigenvalues and Eigenvectors from Linear Algebra
- Find the Eigenvalues and Eigenvectors of an example matrix

LU Decomposition

What is it and why do we need it?

The idea is to decompose a matrix A into an equivalent pair L and U , where L and U are *lower triangular* and *upper triangular* matrices respectively.

$$\begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} = \begin{pmatrix} L_{11} & & \\ L_{21} & L_{22} & \\ L_{31} & L_{32} & L_{33} \end{pmatrix} \begin{pmatrix} U_{11} & U_{12} & U_{13} \\ & U_{22} & U_{23} \\ & & U_{33} \end{pmatrix}$$

- ✓ It helps us *memoize/record* the steps of *Gaussian Elimination*.
- ✓ Triangular system of equations are easy to solve using *Forward Substitution* and *Backward Substitution*.
- ✓ Facilitates efficient solving of $Ax = b$ for multiple column vectors b (especially for large A).

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & -5 & 12 \\ 0 & 2 & -10 \end{bmatrix}$$

$$\xrightarrow[R_2 = -2R_1 + R_2]{E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 6 \\ 0 & 2 & -10 \end{bmatrix}$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

$$R_3 = 2R_2 + R_3$$

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

U

$$A = LU$$

$$E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_{32}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

L

$E_{32} E_{21} A = U$

$\Rightarrow A = \underbrace{E_{21}^{-1} E_{32}^{-1}}_L U$

$\Rightarrow A = LU$

$$A = LU$$

$$PA = LU$$

↙
pivot
matrix

$$A\vec{x} = \vec{b}$$

$$\Rightarrow PA\vec{x} = P\vec{b}$$

$$\Rightarrow LU\vec{x} = P\vec{b}$$

$$\Rightarrow LU\vec{x} = \vec{y}$$

$$\Rightarrow L\vec{y} = \vec{d}$$

✓ Fwd sub. $\rightarrow \vec{y}$

$$U\vec{x} = \vec{y}$$

✓ Bckd sub. $\rightarrow \vec{x}$

$$A(\vec{x}) = \vec{b}$$

(?)

$\vec{b}_1, \vec{b}_2, \dots, \vec{b}_n$

↙

LU Decomposition

An example

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 2 & -5 & 12 \\ 0 & 2 & -10 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & -2 & 3 \\ (2) & -1 & 6 \\ 0 & 2 & -10 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & -2 & 3 \\ (2) & -1 & 6 \\ (0) & (-2) & 2 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 1 & -2 & 3 \\ 0 & -1 & 6 \\ 0 & 0 & 2 \end{pmatrix}$$

$$LU = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 & 3 \\ 0 & -1 & 6 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -2 & 3 \\ 2 & -5 & 12 \\ 0 & 2 & -10 \end{pmatrix} = A.$$

Eigenvalues and Eigenvectors

What are they?

Suppose that A is a square ($n \times n$) matrix. We say that a nonzero vector $\mathbf{v}$ is an eigenvector and a number λ is its eigenvalue if

$$A\mathbf{v} = \lambda\mathbf{v}. \quad (14.1)$$

Geometrically this means that $A\mathbf{v}$ is in the same direction as $\mathbf{v}$, since multiplying a vector by a number changes its length, but not its direction.

$$A\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}$$

$$A\mathbf{v} - \lambda\mathbf{v} = A\mathbf{v} - \lambda I\mathbf{v}$$

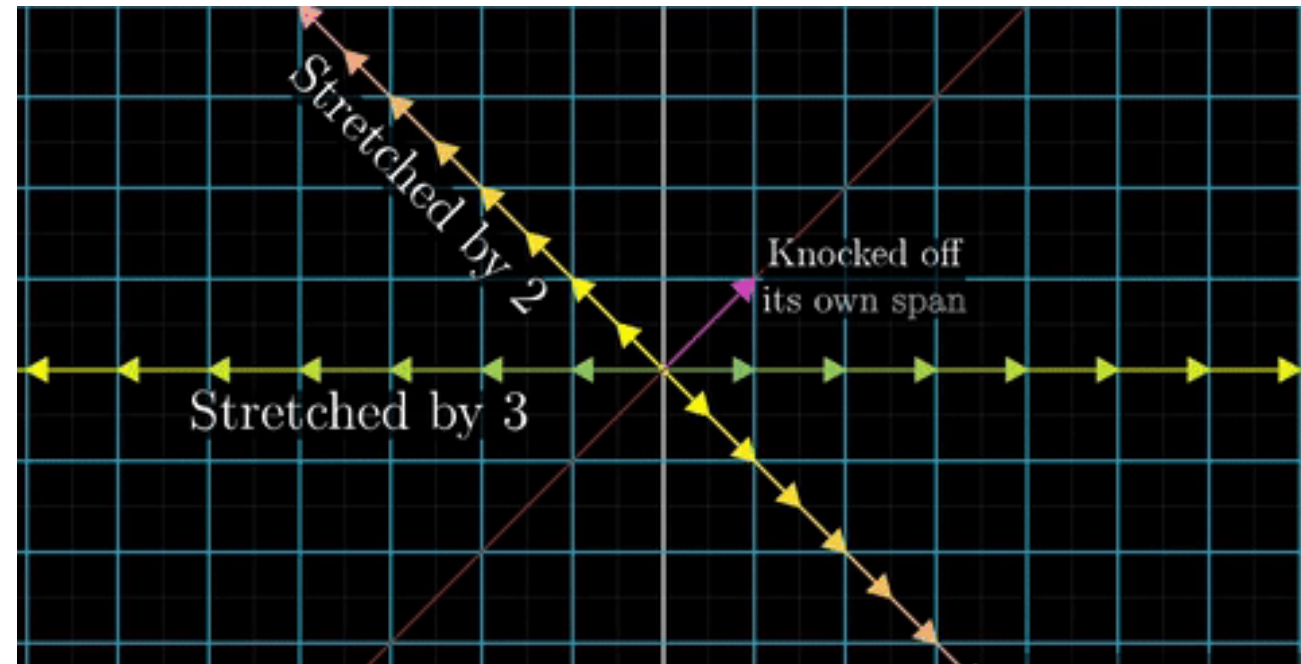
$$= (A - \lambda I)\mathbf{v}$$

$$= \mathbf{0}$$

Must be singular
for non-zero $\vec{v}$.

$$\det(A - \lambda I) = 0.$$

This is called the *characteristic equation*.



Link: <https://youtu.be/PFDu9oVAE-g?si=ARTE9IkmFDfikkrrT>

$$A\vec{v} = \lambda\vec{v}$$

eigenvector

eigenvalue

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A\vec{v} = \lambda I\vec{v}$$

$$\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\Rightarrow A\vec{v} - \lambda I\vec{v} = 0$$

$$\Rightarrow (A - \lambda I)\vec{v} = 0$$

$$\det(A - \lambda I) = 0$$

$$\Rightarrow \underbrace{|A - \lambda I|}_{\text{characteristic eqn}} = 0 \Rightarrow$$

$$\begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} = 0$$

$$A\vec{x} = 0$$

non-zero

$\Downarrow$

singular

$$\det(A) = 0$$

eigen

property
characteristic

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 5 \end{bmatrix}$$

$$(A - \lambda I)\vec{v} = 0$$

$$\Rightarrow |A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 4 \\ 3 & 5-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(5-\lambda) - 3 \times 4 = 0$$

$$\Rightarrow 5 - \lambda - 5\lambda + \lambda^2 - 12 = 0$$

$$\Rightarrow \lambda^2 - 6\lambda - 7 = 0$$

$$\Rightarrow \lambda = 7, -1$$

λ_1 λ_2

for $\lambda_1 = 7$;

$$\begin{bmatrix} 1-7 & 4 \\ 3 & 5-7 \end{bmatrix} \vec{v} = 0$$

$$\Rightarrow \begin{bmatrix} -6 & 4 \\ 3 & -2 \end{bmatrix} \vec{v} = 0$$

$$\Rightarrow \begin{bmatrix} -6 & 4 & | & 0 \\ 3 & -2 & | & 0 \end{bmatrix}$$

$$R_2 = \frac{1}{2}R_1 + R_2$$

$$\begin{bmatrix} -6 & 4 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$-6x + 4y = 0$$

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

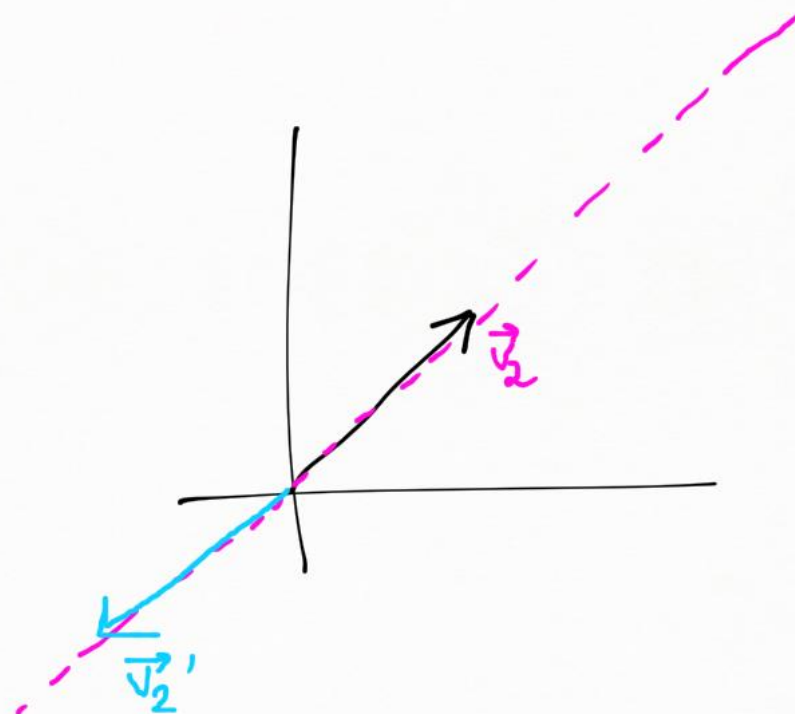
for $\lambda_2 = -1$;

$$\begin{bmatrix} 1+1 & 4 \\ 3 & 5+1 \end{bmatrix} \vec{v} = 0$$

$$\left[\begin{array}{cc|c} 2 & 4 & 0 \\ 3 & 6 & 0 \end{array} \right] \xrightarrow{R_2 = \frac{-3}{2}R_1 + R_2} \left[\begin{array}{cc|c} 2 & 4 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

$$2x + 4y = 0$$

$$\vec{v}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$



Eigenvalues and Eigenvectors

An example

$$A = \begin{pmatrix} 1 & 4 \\ 3 & 5 \end{pmatrix}$$

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 1 & 4 \\ 3 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 4 \\ 3 & 5 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 1 - \lambda & 4 \\ 3 & 5 - \lambda \end{pmatrix}. \end{aligned}$$

The determinant of $A - \lambda I$ is then

$$\begin{aligned} \det(A - \lambda I) &= (1 - \lambda)(5 - \lambda) - 4 \cdot 3 \\ &= -7 - 6\lambda + \lambda^2. \end{aligned}$$

The characteristic equation $\det(A - \lambda I) = 0$ is simply a quadratic equation: $\lambda^2 - 6\lambda - 7 = 0$.

The roots of this equation are $\lambda_1 = 7$ and $\lambda_2 = -1$.

For $\lambda_1 = 7$, the equation for the eigenvector $(A - \lambda I)\mathbf{v} = \mathbf{0}$ is equivalent to the augmented matrix

$$\left(\begin{array}{cc|c} -6 & 4 & 0 \\ 3 & -2 & 0 \end{array} \right)$$

$$3x - 2y = 0$$

we can let $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

This comes from noticing that $(x, y) = (2, 3)$ is a solution of $3x - 2y = 0$.

For $\lambda_2 = -1$,

$$\left(\begin{array}{cc|c} 2 & 4 & 0 \\ 3 & 6 & 0 \end{array} \right) \quad \mathbf{v}_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$A\mathbf{v}_1 = \begin{pmatrix} 1 & 4 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 14 \\ 21 \end{pmatrix} = 7 \begin{pmatrix} 2 \\ 3 \end{pmatrix} = 7\mathbf{v}_1 \quad \text{and}$$

$$A\mathbf{v}_2 = \begin{pmatrix} 1 & 4 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} = -1 \begin{pmatrix} -2 \\ 1 \end{pmatrix} = -1\mathbf{v}_2.$$